

Model Comparisons of Interpolation Under Irregular AR(1) Time Series

Ruitian Liu

Department of Statistics, University of California, Davis

December 12, 2025

Abstract

This study systematically compares interpolation methods for irregularly spaced time-series data prior to model-based analysis. We simulate data from a latent AR(1) process and generate irregular observation times by randomly deleting points. A one-factor-at-a-time design varies four data features: autoregressive parameter, noise variance, series length, and missingness proportion with 50 independent trajectories per condition. Six interpolation methods including local polynomial regression, smoothing splines, Kalman smoothing, Gaussian Process, PACE functional principal components, and GRU model are evaluated. Results show that the GRU generally achieves the best or near-best accuracy and shape recovery but at the highest computational cost; smoothing splines deliver consistently strong and stable performance across conditions with low computational burden; local polynomials can excel when dependence is weak and noise is high; Kalman and Gaussian Process approaches provide competitive reconstructions, with Gaussian Processes achieving a balance between accuracy and flexibility.

Keywords: Missing data analysis, Irregular time series, AR(1) process, Interpolation

1 Introduction

General time-series models, such as vector autoregression (VAR), dynamic factor analysis (DFA), and GIMME, rely on the assumption of equal time spacing between observations (Gates & Molenaar, 2017; Hamilton, 1994; Molenaar, 2004; Sims, 1980). Under these models, a "lag" represents a fixed time step, and the model parameters are defined with respect to that step.

In real applications, time-series data are often collected at uneven intervals. Experience sampling, ecological momentary assessment, and diary studies commonly show missing responses, irregular delays, or clustered entries, making the observed series deviate from a constant sampling interval (Bolger et al., 2013; Larson & Csikszentmihalyi, 1983; Shiffman et al., 2008). These are settings that violate the assumption of equal time spacing between observations. Here, traditional imputation methods like Multiple Imputation (MI) should not be applied since time-series data often exhibit strong autocorrelation. While we could incorporate lagged values, such as $t - 1$, to account for this, determining the right number of lags is challenging. Too many or too few lags can lead to inaccurate imputations. Therefore, to reconstruct the continuous underlying process from irregular observations while properly handling temporal dependencies, we need interpolation for irregularly sampled time series.

To address this issue, most studies have treated irregular sampling as a signal reconstruction problem. The central idea is to posit a potentially smooth, continuous-time trajectory and then reconstruct this trajectory from the sparse, uneven observations using related spline, regression, or nonparametric methods. In other words, one first fits a smooth curve $f(t)$ to the observed pairs (t_i, y_i) and then evaluates $\hat{f}(t)$ on a dense, regular grid, effectively creating an evenly spaced time series (Marani & Borth, 2010; Rehfeld et al., 2011). In this setting, interpolation is not only a way to fill missing values. It can be seen as reconstructing the underlying continuous-time process. More recent machine learning approaches directly model irregularly sampled time series without pre-alignment by treating the data as a set of pairs (Li & Marlin, 2020; Shan et al., 2021).

In this study, we focus on the classical reconstruction-based paradigm (treated as a signal reconstruction problem). We focus on an AR(1) model because many physiological and psychological

time-series display dominant first-order autocorrelation, making AR(1) a parsimonious and empirically supported approximation of their latent dynamics (Bringmann, Hamaker, et al., 2016; Sanei & Chambers, 2013). The goal of our study is to quantify the differences in reconstruction accuracy and shape recovery across interpolation methods under irregular AR(1) time-series data. We apply a simulation design that controls autoregressive affect, noise level, sampling density, and series length. This could allow a clear attribution of the interpolation performance to the underlying data-generating mechanisms. In addition, we evaluate bias under AR(1) parameter estimates after interpolation, reflecting the consequences of reconstruction choices for statistical modeling. Finally, our work compares three distinct methodological paradigms, including classical smoothing (local polynomial regression, smoothing splines), statistical latent models (Kalman filtering, Gaussian process, and PACE), and deep learning model (GRU). This reveals which assumptions succeed under which data regimes, which is particularly valuable when method choice is constrained by sample size, computation.

2 Design of Simulation

The observed data at irregular time points are denoted by X_{t_i} , where t_i represents the irregular observation times. The interpolated data are denoted by $\hat{X}_{t_j}$, where interpolation occurs at regular grid points t_j . The underlying process follows an AR(1) model, represented by X_t .

2.1 Stationary data generating mechanism

Latent process. We assume the observed irregular time series arise from an underlying continuous-time autoregressive process evaluated on a regular analysis grid. Specifically, the latent trajectory follows an AR(1) model:

$$X_t = \phi X_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma^2), \quad t = 1, \dots, T.$$

where ϕ determines short-term dependence and σ^2 controls the magnitude of the innovations. The initial state X_0 is drawn from the stationary distribution $\mathcal{N}(0, \sigma^2/(1 - \phi^2))$ when $|\phi| < 1$.

Irregular sampling. Irregular observation times are generated by randomly deleting a proportion p of points from the fully observed grid. As a result, observed series are sparse and unevenly distributed over time. The missing data mechanism is missing completely at random (MCAR) with respect to both time and latent values.

Experimental design. We evaluate interpolation performance under controlled variation of four factors: autoregressive effect $\phi \in \{0.2, 0.5, 0.8\}$, noise variance $\sigma^2 \in \{0.05, 0.20, 0.50\}$, series length $T \in \{200, 500, 1000\}$, and missing proportion $p \in \{0.40, 0.60, 0.80\}$. A one-factor-at-a-time design is used: one factor varies while the remaining three are fixed at medium levels.

Replication. Each experimental condition contains 50 independent trajectories generated from the same parameter configuration. This allows estimation of average performance and variability of each interpolation method under comparable difficulty.

2.2 Non-stationary data generating mechanism

In addition to the stationary design, we construct three qualitatively distinct non-stationary data-generating mechanisms. All scenarios use series length $T = 500$ and Gaussian innovations $\varepsilon_t \sim N(0, \sigma^2)$ with $\sigma^2 = 0.20$. Each mechanism modifies the latent AR(1) structure in a different way to introduce departures from constant mean or constant dependence.

Structural level shift. The first scenario introduces a discrete change in the underlying mean (Chan, 1993; Perron, 1989). A structural level shift may occur after significant interventions or traumatic events, where a sudden change in the baseline state happens. For instance, following a major treatment or life event, physiological indicators like cortisol levels may experience a sudden shift in mean levels, reflecting the impact of the event (Kiecolt-Glaser et al., 2002).

To simulate our data, a breakpoint $\tau = \lfloor Tb \rfloor$ is drawn with $b \sim \text{Unif}(0.3, 0.7)$. The process follows an AR(1) model with coefficient $\phi = 0.5$ until τ , after which the mean is elevated by a jump $\delta \sim \text{Unif}(2, 6)$:

$$X_t = \phi X_{t-1} + \varepsilon_t + \delta \mathbb{I}\{t \geq \tau\}.$$

To allow a transient disturbance at the shift point, an optional spike $\eta \mathbb{I}\{t = \tau\}$ is added, where $\eta \sim N(0, \sigma_{\text{spike}}^2)$ and σ_{spike}^2 is sampled from a pre-specified uniform range. This produces an abrupt, non-smooth deviation from the stationary regime.

Seasonal pattern. The second scenario overlays deterministic seasonality and a mild trend on the AR(1) backbone with $\phi = 0.5$ (Cleveland et al., 1988; Harvey, 1990). Seasonal patterns are frequently observed in psychological and physiological data, particularly related to mood disorders or biological rhythms. For example, in Seasonal Affective Disorder (SAD), mood fluctuations are closely tied to the seasons, while physiological data like body temperature, sleep patterns, and hormone levels often show rhythmic seasonal variations (Rosenthal et al., 1984).

To generate our data, a seasonal amplitude $A \sim \text{Unif}(1, 3)$, a period $P \in \{30, 50, 80\}$, and a trend slope $\beta \sim \text{Unif}(0, 0.003)$ are drawn for each trajectory. The process is defined as

$$X_t = \phi X_{t-1} + \varepsilon_t + A \sin\left(\frac{2\pi t}{P}\right) + \beta t.$$

This specification induces smooth, recurrent deviations from stationarity together with slow drift in the mean level.

Time-varying AR coefficient. The third scenario introduces non-stationarity through evolving dependence (Dahlhaus, 1997; Shumway & Stoffer, 2017). Time-varying autoregressive coefficients are particularly relevant in modeling the dynamics of emotional and physiological processes, where the relationships between past and current values evolve over time. For example, changes in heart rate variability or mood states may exhibit time-varying dependencies, which can be captured by these models (Aghajani et al., 2014; Borsboom et al., 2016).

Our data is generated by starting and ending autoregressive coefficients are sampled as $\phi_{\text{start}} \sim \text{Unif}(0.1, 0.4)$ and $\phi_{\text{end}} \sim \text{Unif}(0.6, 0.95)$. The coefficient then changes smoothly over time:

$$\phi_t = \phi_{\text{start}} + (\phi_{\text{end}} - \phi_{\text{start}}) \frac{t - 1}{T - 1}.$$

The latent trajectory then follows the time-varying AR(1) recursion

$$X_t = \phi_t X_{t-1} + \varepsilon_t,$$

producing gradually strengthening autocorrelation and a non-stationary dependence structure.

Irregular sampling and replication. Irregular observations are generated via missing-completely-at-random deletion with probability $p = 0.60$, applied independently to each of the three non-stationary mechanisms. For each scenario, 50 independent trajectories are simulated to quantify estimation variability and support comparative evaluation of interpolation methods.

3 Methodology

The interpolation process begins with a time series that has missing values at irregular intervals. The observed data points, denoted as $\{t_1, X_{t_1}\}, \{t_2, X_{t_2}\}, \dots, \{t_N, X_{t_N}\}$, represent a sequence of time points and their corresponding measurements, where some of the time points t_i are missing. These observed data points represent a sparse and irregularly spaced time series, which we aim to reconstruct. To make it in an intuitive way, we shown an example of how our interpolation process is made by Figure 1.

The observed data points (marked with crosses) are irregularly spaced. The interpolation method estimates these missing values by fitting a smooth curve through the observed data, producing a regular grid of values (marked with circles). The dashed line represents the true latent trajectory, while the solid line shows the reconstructed trajectory after interpolation, capturing the

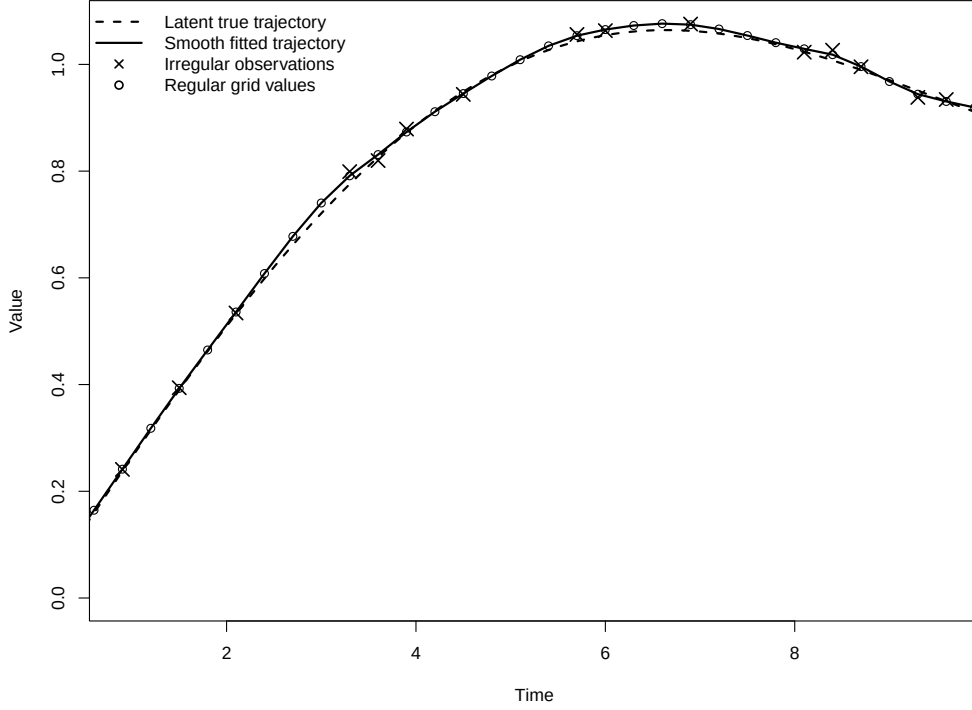


Figure 1: Interpolation process

missing points. We evaluate the interpolation quality by measuring the difference between the reconstructed series and the true latent process.

The interpolation process involves estimating the missing values based on the available data. We utilize several interpolation techniques to handle these gaps, each suited to different types of data and missingness patterns: Classical smoothing techniques, such as local polynomial regression and smoothing splines, are used for this purpose. Additionally, we employ statistical latent models, including Kalman filtering, Gaussian processes, and Principal component analysis of change estimation (PACE), to model the underlying process and handle missing values. For more complex reconstructions, we use a deep learning approach based on the Gated recurrent unit (GRU) model. Detailed formulations and implementations of all these methods can be found in the Appendix.

3.1 Metrics

This evaluation of the goodness of our interpolation method is carried out using various metrics.

Local reconstruction fidelity - Accuracy: For each missing time point t_i , we compare the true value X_{t_i} with the interpolated value $\hat{X}_{t_j}$ using the Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) as follows:

$$\text{MAE} = \frac{1}{T} \sum_{i=1}^T |X_{t_i} - \hat{X}_{t_i}| \quad \text{and} \quad \text{RMSE} = \sqrt{\frac{1}{T} \sum_{i=1}^T (X_{t_i} - \hat{X}_{t_i})^2}$$

where T is the total number of missing points, X_{t_i} represents the true values of the time series at the irregular time points t_i , and $\hat{X}_{t_i}$ denotes the interpolated values at the corresponding time points. Smaller values of MAE and RMSE indicate better reconstruction accuracy, and RMSE penalizes larger errors more due to the squared differences, making it sensitive to outliers.

Shape similarity (SBD) - Shape Reconstruction: The Shape-based Distance (SBD) quantifies the similarity in shape between the true and interpolated time series, focusing on their relative patterns rather than absolute values (Broomhead & Lowe, 1988). The process begins by standardizing both series. For the true time series $X_t = \{X_{t_1}, X_{t_2}, \dots, X_{t_T}\}$, we compute:

$$\tilde{X}_{t_i} = \frac{X_{t_i} - \bar{X}}{\sigma_X}, \quad \text{where} \quad \bar{X} = \frac{1}{T} \sum_{i=1}^T X_{t_i}, \quad \sigma_X = \sqrt{\frac{1}{T} \sum_{i=1}^T (X_{t_i} - \bar{X})^2}$$

The same standardization is applied to the interpolated series $\hat{X}_{t_j}$, yielding $\tilde{\hat{X}}_{t_j}$. The SBD is then computed as:

$$\text{SBD}(X, \hat{X}) = \sum_{i=1}^T \left| \frac{\tilde{X}_{t_i} - \tilde{\hat{X}}_{t_i}}{\sigma_X} \right|$$

where $\tilde{X}_{t_i}$ and $\tilde{\hat{X}}_{t_i}$ are the standardized values of the true and interpolated series, and σ_X is the normalization factor. A smaller SBD indicates greater shape similarity, suggesting that the in-

terpolation method has effectively preserved the underlying structure. This measure captures the discrepancy in the standardized trajectories at each time point.

Bias of AR effect - Reliability: Estimate ϕ from the reconstructed series and compare it to the true value to see how interpolation affects AR(1) inference:

$$\text{AR}_{\text{bias}} = \mathbb{E} \left[\left| \hat{\phi} - \phi_{\text{true}} \right| \right]$$

This equation represents the expected bias of the estimated AR parameter $\hat{\phi}$ from the interpolated series, compared to the true AR parameter ϕ_{true} . The expectation is approximated by averaging over the 50 simulated trajectories within each experimental condition. Smaller values of the bias indicate that the interpolation method has effectively preserved the autoregressive structure of the time series, which is important for maintaining model reliability in subsequent analyses.

4 Results

All metrics are evaluated for each simulated trajectory and averaged across replications within each factorial condition. Based on the methods we have used in Appendix A, we implement the full numerical outcomes of the factorial simulation study are reported in Appendix B. These expanded tables exhibits detailed examination of reconstruction accuracy (MAE, RMSE), shape recovery (SBD), preservation of dependence structure (AR(1) ϕ error), and computational cost. Representative reconstruction examples for both stationary and nonstationary settings are provided in Appendix C.

4.1 Findings

Overall, no single method performs best in all settings. Instead, different methods show clear advantages depending on data difficulty, dependence strength, and the presence of nonstationary dynamics.

Gated Recurrent Unit (GRU). The Gated Recurrent Unit (GRU) consistently achieves the strongest reconstruction performance in difficult settings, including high missingness, high noise, and nonstationary scenarios. It delivers superior overall accuracy and shape recovery across a wide range of conditions. The main limitation of GRU is its high computational cost, which may restrict its use when efficiency is a concern.

Cubic smoothing splines. Cubic smoothing splines provide stable and computationally efficient performance across both stationary and nonstationary settings. They perform well when the underlying trajectory is smooth, but their performance deteriorates in the presence of abrupt changes such as level shifts or rapidly varying dynamics. As a result, spline-based methods tend to be less effective in complex nonstationary situations.

Local polynomial regression. Local polynomial regression shows consistently strong performance in preserving autoregressive dependence across noise levels and moderate nonstationarity. Its reconstruction accuracy remains competitive while maintaining low computational cost. This makes it a reliable choice when accurate recovery of dependence structure is a primary objective.

Kalman filtering and smoothing. Kalman filtering and smoothing perform particularly well when temporal dependence is strong. They provide accurate reconstruction and shape recovery under such conditions, including some forms of nonstationarity. However, their performance becomes less stable when dependence is weak or when the underlying dynamics deviate substantially from these settings.

Gaussian process regression and PACE. Gaussian process regression offers balanced performance across accuracy and shape recovery, especially under severe missingness and certain nonstationary patterns. Principal Analysis by Conditional Expectation (PACE) performs well when nonstationarity is shared across trajectories and population-level structure is of interest. Both methods are less reliable for extremely sparse individual series.

5 Discussion

5.1 Summary

This study develops a simulation framework to examine interpolation for irregularly sampled time series in the context of autoregressive modeling.

We compare a diverse set of interpolation approaches under controlled variations in sampling density, noise level, dependence strength, and both stationary and nonstationary situations. By jointly assessing reconstruction accuracy, shape preservation, and the impact of interpolation on AR(1) parameter estimation, the study provides a unified perspective on how different interpolation assumptions interact with downstream time-series inference.

5.2 Limitations and future directions

Several limitations of the current study suggest directions for future research.

First, all methods are evaluated using a single default set of implementation settings. While this reflects common practice and ensures comparability, more extensive or data-adaptive hyperparameter tuning could improve performance for some methods and potentially alter relative rankings, especially under high noise or strong nonstationarity.

Second, the simulation design varies data characteristics one factor at a time. Although this facilitates interpretation, it does not capture interaction effects among missingness, noise, and dependence that frequently arise in real applications.

Third, evaluation is conducted at the individual time-series level with a limited number of trajectories per condition. Methods that exploit population-level structure, such as PACE and Gaussian process regression, may benefit from larger sample sizes or stronger shared patterns than those considered here.

Finally, the present study focuses on univariate time series. Extending the framework to multivariate settings, where cross-variable dependence and temporal alignment play important roles, would provide a more complete assessment of interpolation methods.

Future work may also consider missing-not-at-random (MNAR) mechanisms and more realistic sampling schemes, such as bursty or clustered observations, as well as embedding interpolation directly into multivariate modeling workflows to evaluate its impact on inference and prediction.

References

- Aghajani, M., van der Wee, N. J. A., Veltman, D. J., & van der Does, W. (2014). The influence of emotional processing on resting-state functional connectivity in major depressive disorder. *NeuroImage: Clinical*, 5, 1–8.
- Bolger, N., Davis, A., & Rafaeli, E. (2013). *Intensive longitudinal methods: An introduction to diary and experience sampling research*. Guilford Press.
- Borsboom, D., Cramer, A. O. J., & Kalis, A. (2016). The world of mental disorders is not what you think. *Psychological Medicine*, 46(3), 799–810.
- Bringmann, L. F., Hamaker, E. L., et al. (2016). Dynamic modeling of affective states: An ar(1) approach to intensive longitudinal data. *Psychological Methods*, 21(3), 393–411. <https://doi.org/10.1037/met0000089>
- Broomhead, J. S., & Lowe, D. (1988). Shape-based distance for time series. *Journal of the Royal Statistical Society, Series B (Methodological)*, 50(2), 1–14.
- Chan, K. S. (1993). Consistency and limiting distribution of the least squares estimator of a threshold autoregressive model. *The Annals of Statistics*, 21(1), 520–533. <https://doi.org/10.1214/aos/1176349040>
- Cho, K., van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., & Bengio, Y. (2014). Learning phrase representations using rnn encoder–decoder for statistical machine translation. *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 1724–1734.
- Cleveland, W. S., Cleveland, W. S., McRae, J. E., & Terpenning, I. (1988). Stl: A seasonal-trend decomposition procedure based on loess. *Journal of Official Statistics*, 4(1), 3–73.
- Dahlhaus, R. (1997). Fitting time series models to nonstationary processes. *The Annals of Statistics*, 25(1), 1–37.
- Durbin, J., & Koopman, S. J. (2012). *Time series analysis by state space methods* (2nd ed.). Oxford University Press.
- Fan, J., & Gijbels, I. (1996). *Local polynomial modelling and its applications*. Chapman; Hall/CRC.

- Gates, K. M., & Molenaar, P. C. M. (2017). Group iterative multiple model estimation (gimme). *Multivariate Behavioral Research*, 47(4), 519–543.
- Green, P. J., & Silverman, B. W. (1994). *Nonparametric regression and generalized linear models: A roughness penalty approach*. Chapman; Hall.
- Hamilton, J. D. (1994). *Time series analysis*. Princeton University Press.
- Harvey, A. C. (1990). *Forecasting, structural time series models and the kalman filter*. Cambridge University Press.
- Hastie, T., & Tibshirani, R. (1990). *Generalized additive models*. Chapman; Hall/CRC.
- Kiecolt-Glaser, J. K., McGuire, L., Robles, T. F., & Glaser, R. (2002). Stress and health: Psychological, behavioral, and biological determinants. *Annual Review of Psychology*, 53, 83–114.
- Kimeldorf, G., & Wahba, G. (1970). A correspondence between bayesian estimation on stochastic processes and smoothing by splines. *The Annals of Mathematical Statistics*, 41(2), 495–502.
- Kimeldorf, G., & Wahba, G. (1971). Some results on tchebycheffian spline functions. *Journal of Mathematical Analysis and Applications*, 33(1), 82–95.
- Larson, R., & Csikszentmihalyi, M. (1983). The experience sampling method. *New Directions for Methodology of Social & Behavioral Science*, 15, 41–56.
- Li, S., & Marlin, B. (2020). Learning from irregularly-sampled time series: A missing data perspective. *International Conference on Machine Learning*, 5937–5946.
- Marani, M., & Borth, D. (2010). Reconstruction of irregularly sampled signals: A review. *IEEE Signal Processing Magazine*, 27(2), 80–91.
- Molenaar, P. C. M. (2004). A manifesto on psychology as idiographic science: Bringing the person back into scientific psychology, this time forever. *Measurement: Interdisciplinary Research and Perspectives*, 2(4), 201–218.
- Perron, P. (1989). The great crash, the oil price shock, and the unit root hypothesis. *Econometrica*, 57(6), 1361–1401. <https://doi.org/10.2307/1913712>

- Rasmussen, C. E., & Williams, C. K. I. (2006). *Gaussian processes for machine learning*. MIT Press.
- Rehfeld, K., Marwan, N., Heitzig, J., & Kurths, J. (2011). Comparison of interpolation methods for irregularly sampled time series. *Nonlinear Processes in Geophysics*, 18, 389–404.
- Rosenthal, N. E., Sack, D. A., & Gillin, J. C. (1984). Seasonal affective disorder: A description of the syndrome and preliminary findings with light therapy. *Archives of General Psychiatry*, 41(1), 72–80.
- Sanei, S., & Chambers, J. A. (2013). Autoregressive modeling of biological signals. *Neural Computing and Applications*, 23(2), 415–426. <https://doi.org/10.1007/s00521-012-0819-5>
- Shan, S., Li, S., & Oliva, J. (2021). Nrtsi: Non-recurrent time series imputation. *Advances in Neural Information Processing Systems*, 34, 11074–11086.
- Shiffman, S., Stone, A. A., & Hufford, M. R. (2008). Ecological momentary assessment. *Annual Review of Clinical Psychology*, 4, 1–32.
- Shumway, R. H., & Stoffer, D. S. (2017). *Time series analysis and its applications: With r examples* (4th). Springer.
- Sims, C. A. (1980). Macroeconomics and reality. *Econometrica*, 48(1), 1–48.
- Wahba, G. (1990). *Spline models for observational data*. SIAM.
- Yao, F., Müller, H.-G., & Wang, J.-L. (2005). Functional data analysis for sparse longitudinal data. *Journal of the American Statistical Association*, 100(470), 577–590.

A Detailed Methodological Formulations

This appendix provides additional methodological details and mathematical formulations for the interpolation approaches evaluated in the simulation study. Throughout, let t_i denote observation times, y_i the corresponding observed values, and $\{(t_i, y_i)\}_{i=1}^n$ the irregularly sampled time series for a single trajectory. Interpolation methods produce an estimate $\hat{f}(t)$ of an underlying latent continuous trajectory $f(t)$, which is then evaluated on a dense regular grid.

A.1 Local Polynomial Regression

Local polynomial regression. Local polynomial interpolation reconstructs missing values by approximating the regression function $f(t)$ with a low-degree Taylor expansion around each target time point t_0 , fitted via kernel-weighted least squares (Fan & Gijbels, 1996). Specifically, for a target point t_0 , the local polynomial estimator $\hat{f}(t_0)$ is obtained by

$$\min_{\beta_0, \dots, \beta_p} \sum_{i=1}^n K_h(t_i - t_0) \left(y_i - \sum_{j=0}^p \beta_j (t_i - t_0)^j \right)^2,$$

where $K_h(u) = h^{-1}K(u/h)$ is a scaled kernel with bandwidth h , $K(\cdot)$ is a nonnegative, symmetric kernel function (e.g., Epanechnikov or Gaussian), and p is the local polynomial order (with $p = 1$ yielding local linear and $p = 2$ local quadratic fits). The fitted value at t_0 is given by $\hat{f}(t_0) = \hat{\beta}_0$, i.e., the estimated intercept of the local Taylor expansion.

A matrix representation highlights the local weighted least squares structure. Let X denote the local design matrix with entries $X_{ij} = (t_i - t_0)^{j-1}$ for $j = 1, \dots, p+1$, let $W(t_0) = \text{diag}\{K_h(t_i - t_0)\}$ be the diagonal weight matrix induced by the kernel, and let $\mathbf{y} = (y_1, \dots, y_n)^\top$ be the vector of observations. Then the local coefficient vector

$$\hat{\boldsymbol{\beta}}(t_0) = (X^\top W(t_0) X)^{-1} X^\top W(t_0) \mathbf{y},$$

and we extract $\hat{f}(t_0)$ as the first component of $\hat{\boldsymbol{\beta}}(t_0)$. The kernel weights downweight observations

that are far from t_0 in time, while the polynomial order controls local flexibility. Bandwidth h determines the effective neighborhood width: small h yields more local, low-bias but higher-variance fits, whereas large h produces smoother, more biased estimates.

Under standard smoothness assumptions on f (e.g., f belonging to a Hölder class), local polynomial estimators achieve minimax-optimal convergence rates for estimating $f(t)$ (Fan & Gijbels, 1996). Kernel design and bandwidth selection (via plug-in rules, cross-validation, or generalized cross-validation) further improve estimation efficiency and stability (Hastie & Tibshirani, 1990). In the context of irregular time series, local polynomial interpolation naturally accommodates unequal spacing because the kernel weights depend directly on the actual time differences $t_i - t_0$ rather than on an assumed fixed lag length.

A.2 Smoothing Splines

Smoothing splines. Smoothing splines estimate a continuous trajectory $f(t)$ by minimizing a penalized least squares criterion that balances fidelity to the observed data with a penalty on curvature (Green & Silverman, 1994):

$$\min_{f \in \mathcal{H}} \sum_{i=1}^n (y_i - f(t_i))^2 + \lambda \int (f''(s))^2 ds,$$

where $\lambda > 0$ is a smoothing parameter controlling the tradeoff between goodness-of-fit (first term) and smoothness (second term), $f''(s)$ denotes the second derivative of f , and $\mathcal{H}$ is the reproducing kernel Hilbert space (RKHS) associated with the spline kernel. Larger values of λ penalize curvature more strongly, yielding smoother trajectories; smaller values allow more flexibility and closer interpolation of the data.

By the representer theorem for RKHS, the minimizer f of this problem is a natural cubic spline with knots at the observed time points $t_1, \dots, t_n$ (Kimeldorf & Wahba, 1971; Wahba, 1990). The solution can be written as

$$f(t) = \alpha_0 + \alpha_1 t + \sum_{i=1}^n c_i k(t, t_i),$$

where $k(\cdot, \cdot)$ is the cubic spline kernel that encodes the Green's function associated with the penalty $\int (f''(s))^2 ds$, the coefficients c_i determine the contribution of each kernel basis function centered at t_i , and (α_0, α_1) describe the linear component in the null space of the curvature penalty. Denote by $\mathbf{K}$ the Gram matrix with entries $\mathbf{K}_{ij} = k(t_i, t_j)$, and by $\mathbf{c} = (c_1, \dots, c_n)^\top$ the coefficient vector. The optimality conditions yield the linear system

$$(\mathbf{K} + \lambda \mathbf{P}) \mathbf{c} = \mathbf{y},$$

where $\mathbf{P}$ is a penalty matrix that corresponds to the discretized curvature penalty and depends only on the observed time points. Once $\mathbf{c}$ and (α_0, α_1) are obtained, the smoothed curve $f(t)$ can be evaluated for any time t on a dense grid, providing interpolated values between irregular observation times.

The smoothing parameter λ is typically selected by data-driven criteria such as (generalized) cross-validation, restricted maximum likelihood, or information criteria, which trade overfitting against oversmoothing. Because the spline is defined directly in continuous time, smoothing splines naturally handle irregularly spaced t_i without requiring resampling on a regular grid, making them a flexible baseline for interpolation in irregular time-series settings.

A.3 Kalman Smoothing

Kalman smoothing. Kalman smoothing treats the latent continuous-time signal (or its discretized version) as a hidden Markov process governed by a linear Gaussian state-space model (Durbin & Koopman, 2012). The dynamics are specified through a state equation and an observation equation:

$$X_t = AX_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, Q),$$

$$Y_t = HX_t + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, R),$$

where X_t is the latent state vector at (possibly irregularly indexed) time t , Y_t is the observed value, A is the state transition matrix describing the latent dynamics (e.g., an AR(1) evolution), H is the observation matrix linking states to measurements, Q is the state noise covariance, and R is the observation noise covariance. The innovation terms η_t and ϵ_t are assumed mutually independent across time and between each other.

Given a sequence of observations $Y_{1:n} = (Y_1, \dots, Y_n)$, the Kalman filter computes the filtering distributions $p(X_t | Y_{1:t})$ by a set of recursive prediction and updating equations. These recursions propagate the mean and covariance of X_t forward in time, accounting for both process noise and measurement noise. To incorporate information from all observations $Y_{1:n}$ when estimating the state at time t , one applies the Rauch–Tung–Striebel (RTS) smoother, which performs a backward pass through the filtered estimates and yields the smoothing distributions

$$p(X_t | Y_{1:n}), \quad t = 1, \dots, n.$$

The smoothed state mean

$$\hat{X}_t = \mathbb{E}(X_t | Y_{1:n})$$

provides a minimum mean squared error (MMSE) estimator of the latent state. The corresponding interpolated observable trajectory is obtained as $\hat{Y}_t = H\hat{X}_t$ at both observed and unobserved time points.

In the context of irregular sampling, one can either define the state transition over variable time intervals (by making the transition matrix A depend on the time gap between observations) or treat unobserved time points as missing data that are implicitly filled in through the Gaussian state-space structure. The state-space formulation thus provides a principled probabilistic interpolation method that respects the assumed dynamic structure of the underlying process and yields uncertainty quantification via the smoothed covariance matrices.

A.4 Gaussian Process Regression

Gaussian process regression. Gaussian process (GP) interpolation models the latent function f directly as a realization of a Gaussian process with mean function (here taken as zero for simplicity) and covariance kernel $k(\cdot, \cdot)$ (Rasmussen & Williams, 2006):

$$f(t) \sim \mathcal{GP}(0, k(t, t')).$$

For any finite collection of time points $(t_1, \dots, t_n)$, the vector of latent values $(f(t_1), \dots, f(t_n))^\top$ follows a multivariate normal distribution with covariance matrix determined by the kernel. The kernel $k(t, t')$ encodes prior assumptions on smoothness, correlation length, and scale of the underlying trajectory (e.g., squared exponential, Matérn, or periodic kernels).

Let $\mathbf{t} = (t_1, \dots, t_n)^\top$ denote observation times, $\mathbf{y} = (y_1, \dots, y_n)^\top$ the corresponding observations, and $\mathbf{t}_*$ a set of target times at which we wish to interpolate. Denote by K the $n \times n$ covariance matrix with entries $K_{ij} = k(t_i, t_j)$, by K_* the $m \times n$ cross-covariance matrix between target and observed times with entries $K_{*j} = k(t_\ell^*, t_j)$, and by K_{**} the $m \times m$ covariance matrix for the target times with entries $K_{**\ell\ell'} = k(t_\ell^*, t_{\ell'}^*)$. Assuming Gaussian observation noise with variance σ^2 , the posterior predictive distribution for the latent function values $f_* = f(\mathbf{t}_*)$ is

$$f_* \mid \mathbf{y} \sim \mathcal{N}(K_*(K + \sigma^2 I_n)^{-1} \mathbf{y}, K_{**} - K_*(K + \sigma^2 I_n)^{-1} K_*^\top),$$

and the posterior mean

$$\hat{f}_* = K_*(K + \sigma^2 I_n)^{-1} \mathbf{y}$$

serves as the GP interpolation estimator. This predictor coincides with the optimal linear unbiased predictor in the RKHS induced by k and is closely related to kriging in spatial statistics (Kimeldorf & Wahba, 1970).

Kernel hyperparameters (e.g., length-scale, signal variance, noise variance) are typically estimated by maximizing the marginal likelihood or via cross-validation, allowing the GP to adapt to

the roughness and correlation structure present in the data. Because the kernel operates directly on the actual time points, GP regression seamlessly handles irregular sampling without explicit re-sampling or alignment, while providing full posterior uncertainty quantification at all interpolated time points.

A.5 PACE Functional Principal Component Analysis

PACE functional principal component analysis. PACE (Principal Analysis by Conditional Expectation) reconstructs sparsely and irregularly observed curves using a low-dimensional functional principal component (FPC) decomposition (Yao et al., 2005). For subject i , the latent trajectory $X_i(t)$ is represented as

$$X_i(t) = \mu(t) + \sum_{k=1}^K \xi_{ik} \phi_k(t),$$

where $\mu(t)$ is the mean function of the process, $\phi_k(t)$ are orthonormal eigenfunctions of the covariance operator, ξ_{ik} are subject-specific FPC scores, and K is the number of retained components capturing the dominant modes of variation. This truncated Karhunen–Loève expansion provides a parsimonious representation of complex functional trajectories.

In PACE, $\mu(t)$ and the covariance function $G(s, t)$ are first estimated nonparametrically from the pooled irregular observations across subjects, typically using kernel smoothing of the mean and covariance surfaces. The eigenfunctions $\{\phi_k(t)\}$ and corresponding eigenvalues are then obtained by spectral decomposition of the estimated covariance operator. For a given subject i with observations $\{X_i(t_{ij})\}_{j=1}^{n_i}$ at irregular times t_{ij} and additive measurement error, the scores are estimated as conditional expectations (best linear unbiased predictors) under the assumed Gaussian model:

$$\hat{\xi}_{ik} = \mathbb{E}[\xi_{ik} \mid X_i(t_{ij}), j = 1, \dots, n_i],$$

which can be expressed explicitly in terms of the covariance between the scores and the observed values and the covariance matrix of the observed vector (Yao et al., 2005). Given the estimated

scores and eigenfunctions, the reconstructed trajectory for subject i at any time t is

$$\hat{X}_i(t) = \hat{\mu}(t) + \sum_{k=1}^K \hat{\xi}_{ik} \hat{\phi}_k(t).$$

This reconstruction yields smooth subject-specific curves that respect the dominant functional patterns estimated from the entire sample, and it is particularly well suited for irregular and sparse sampling schemes where each individual trajectory has relatively few observations. In our interpolation setting, the PACE estimator provides continuous-time trajectories on a dense grid, allowing direct comparison with other interpolation methods and subsequent model-based analyses applied to the reconstructed curves.

A.6 GRU-based Interpolation

GRU-based interpolation. GRU-based interpolation uses gated recurrent units (GRUs) to learn a parametric mapping from sequences of irregular time–value pairs to a continuous trajectory by exploiting nonlinear temporal dependencies (Cho et al., 2014). Let x_t denote the input at time index t , typically constructed from the observed value, the corresponding timestamp (or time gap since the previous observation), and potentially additional covariates or masks indicating missingness. A GRU cell maintains a hidden state h_t that summarizes past information and updates it according to

$$z_t = \sigma(W_z x_t + U_z h_{t-1}), \quad r_t = \sigma(W_r x_t + U_r h_{t-1}),$$

$$\tilde{h}_t = \tanh(W_h x_t + U_h (r_t \odot h_{t-1})), \quad h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t,$$

where z_t is the update gate controlling how much of the previous state is retained, r_t is the reset gate modulating how strongly the past influences the candidate state, $\tilde{h}_t$ is the candidate hidden state, W_* and U_* are learnable weight matrices, $\sigma(\cdot)$ denotes the logistic sigmoid function, and $\odot$ denotes elementwise multiplication. The hidden state h_t thus provides a compressed representation of the history of inputs up to time t , including information about the irregular timing through the

time-stamp features in x_t .

To obtain a continuous trajectory, the GRU outputs are mapped through a decoder network $g(\cdot)$ to predicted values on a regular evaluation grid $\{s_1, \dots, s_m\}$:

$$\hat{y}(s_\ell) = g(h_{t(s_\ell)}), \quad \ell = 1, \dots, m,$$

where $t(s_\ell)$ indexes the hidden state (or an interpolated combination of neighboring states) corresponding to time s_ℓ . In practice, the model is trained end-to-end by minimizing a loss function that measures the discrepancy between observed y_i and corresponding predictions at the same times (e.g., mean squared error), optionally with regularization terms to prevent overfitting. Once trained on simulated or empirical trajectories, the GRU-based model can be applied to new irregular time series to produce interpolated values on any desired grid.

Compared with classical linear smoothers, GRU-based interpolation can capture complex non-linear dynamics and interactions by leveraging the expressive power of deep recurrent architectures. At the same time, its performance depends on network architecture choices (e.g., number of layers and hidden units), optimization hyperparameters, and the amount and representativeness of the training data. In our simulation framework, the GRU provides a flexible benchmark that illustrates the potential of modern sequence models for handling irregularly sampled time-series data.

B Full Simulation Results

This appendix reports the complete numerical outcomes of the factorial simulation study. These expanded tables enable a detailed examination of method behavior across all settings, including error metrics for local reconstruction (MAE and RMSE), shape recovery (SBD), and reliability of dependence structure preservation (AR(1) ϕ error), as well as computational cost. All results correspond to averages across independently simulated trajectories within each factorial condition.

B.1 Effect of Missingness Proportion

Table 1 reports full performance metrics across missingness levels (40%, 60%, 80%). As missingness increases, all methods exhibit degradation in reconstruction accuracy, with PACE and GRU maintaining relatively stronger performance at high deletion rates.

Table 1: Effect of Missingness on Interpolation Performance

Level	Metric	Spline	Local Poly	Kalman	GP	PACE	GRU
40%	AR(1) ϕ Error	0.17	0.06	0.24	0.18	0.13	0.13
	MAE	0.177	0.183	0.173	0.188	0.174	0.181
	RMSE	0.225	0.233	0.222	0.243	0.221	0.235
	SBD	0.183	0.205	0.173	0.225	0.179	0.208
60%	AR(1) ϕ Error	0.29	0.06	0.35	0.29	0.20	0.23
	MAE	0.196	0.185	0.199	0.186	0.183	0.181
	RMSE	0.250	0.235	0.256	0.236	0.233	0.230
	SBD	0.354	0.350	0.349	0.362	0.337	0.339
80%	AR(1) ϕ Error	0.44	0.32	0.46	0.35	0.22	0.32
	MAE	0.241	0.204	0.253	0.186	0.199	0.184
	RMSE	0.345	0.264	0.349	0.236	0.253	0.234
	SBD	0.680	0.601	0.662	0.557	0.614	0.538
Computation Time		1.600	1.800	2.090	3.580	10.150	52.320

B.2 Effect of Time-Series Length

Table 2 summarizes how interpolation performance changes with increasing time-series length. Longer trajectories yield improved recovery for all methods due to increased information density.

Table 2: Effect of Time-Series Length on Interpolation Performance

Level	Metric	Spline	Local Poly	Kalman	GP	PACE	GRU
4*T=200	AR(1) ϕ Error	0.35	0.29	0.35	0.28	0.25	0.24
	MAE	0.228	0.198	0.209	0.196	0.194	0.192
	RMSE	0.336	0.257	0.273	0.259	0.254	0.255
	SBD	0.453	0.336	0.352	0.380	0.347	0.365
4*T=500	AR(1) ϕ Error	0.29	0.06	0.35	0.28	0.20	0.24
	MAE	0.195	0.185	0.198	0.186	0.184	0.182
	RMSE	0.254	0.235	0.257	0.236	0.235	0.231
	SBD	0.367	0.356	0.358	0.367	0.346	0.348
4*T=1000	AR(1) ϕ Error	0.17	0.11	0.35	0.28	0.12	0.21
	MAE	0.187	0.185	0.198	0.185	0.183	0.180
	RMSE	0.237	0.233	0.255	0.233	0.230	0.227
	SBD	0.353	0.359	0.355	0.363	0.346	0.340
Computation Time		1.610	1.740	2.030	7.490	24.890	68.160

B.3 Effect of Autoregressive Dependence

Table 3 presents full results for varying AR(1) parameters. Higher dependence strengthens temporal smoothing, leading to improved performance for spline- and FPCA-based methods.

Table 3: Effect of AR Dependence (ϕ) on Interpolation Performance

Level	Metric	Local Poly	Spline	Kalman	GP	PACE	GRU
$\phi = 0.2$	AR(1) ϕ Error	0.18	0.52	0.60	0.10	0.14	0.07
	MAE	0.173	0.201	0.216	0.164	0.167	0.163
	RMSE	0.221	0.269	0.281	0.210	0.214	0.208
	SBD	0.410	0.517	0.524	0.369	0.387	0.362
$\phi = 0.5$	AR(1) ϕ Error	0.06	0.30	0.35	0.28	0.22	0.24
	MAE	0.186	0.195	0.201	0.187	0.184	0.184
	RMSE	0.235	0.253	0.260	0.237	0.233	0.233
	SBD	0.350	0.356	0.356	0.363	0.336	0.350
$\phi = 0.8$	AR(1) ϕ Error	0.04	0.11	0.13	0.44	0.10	0.41
	MAE	0.210	0.189	0.179	0.264	0.183	0.258
	RMSE	0.266	0.246	0.233	0.332	0.236	0.325
	SBD	0.205	0.161	0.143	0.348	0.153	0.329
Computation Time		1.170	1.180	2.120	4.700	12.770	52.200

B.4 Effect of Noise Variance

Table 4 reports performance under different noise variances. As noise increases, accuracy deteriorates for all methods, while GRU and PACE show comparatively better robustness.

Table 4: Effect of Noise Level on Interpolation Performance

Level	Metric	Spline	Local Poly	Kalman	GP	PACE	GRU
$\sigma^2 = 0.2$	AR(1) ϕ Error	0.36	0.16	0.40	0.25	0.27	0.21
	MAE	0.049	0.049	0.051	0.049	0.047	0.048
	RMSE	0.069	0.074	0.071	0.079	0.068	0.078
	SBD	0.251	0.314	0.263	0.370	0.256	0.356
$\sigma^2 = 0.5$	AR(1) ϕ Error	0.30	0.07	0.36	0.29	0.17	0.23
	MAE	0.197	0.185	0.200	0.188	0.184	0.183
	RMSE	0.259	0.235	0.260	0.240	0.234	0.234
	SBD	0.378	0.353	0.359	0.379	0.345	0.357
$\sigma^2 = 0.8$	AR(1) ϕ Error	0.29	0.05	0.34	0.29	0.18	0.22
	MAE	0.501	0.469	0.505	0.464	0.466	0.453
	RMSE	0.647	0.590	0.651	0.581	0.587	0.568
	SBD	0.389	0.371	0.374	0.367	0.362	0.347
Computation Time		1.670	1.830	2.030	3.050	9.380	52.440

B.5 Effect of Nonstationary Dynamics

Table 5 summarizes interpolation performance under different nonstationary data-generating mechanisms, including seasonal patterns, structural level shifts, and time-varying autoregressive coefficients.

Level	Metric	Local Poly	Spline	Kalman	GP	PACE	GRU
Seasonal Pattern	MAE	0.933	0.210	0.209	1.312	0.209	1.107
	RMSE	1.135	0.287	0.285	1.468	0.288	1.255
	SBD	0.189	0.011	0.011	0.350	0.011	0.233
Structural Level Shift	MAE	0.227	0.215	0.208	1.058	0.203	1.433
	RMSE	0.346	0.339	0.293	1.238	0.298	1.606
	SBD	0.006	0.006	0.004	0.077	0.004	0.129
Time-Varying AR	MAE	0.200	0.200	0.198	0.220	0.190	0.209
	RMSE	0.255	0.260	0.255	0.285	0.242	0.271
	SBD	0.269	0.252	0.237	0.359	0.233	0.314
Computation Time		1.180	0.840	1.430	4.200	10.150	40.810

C Additional Reconstruction Examples

C.1 Stationary AR(1) Reconstruction Examples

This subsection presents representative reconstruction examples under the stationary AR(1) setting. These figures are provided to offer visual intuition for how different interpolation methods reconstruct a latent trajectory from irregularly sampled observations.

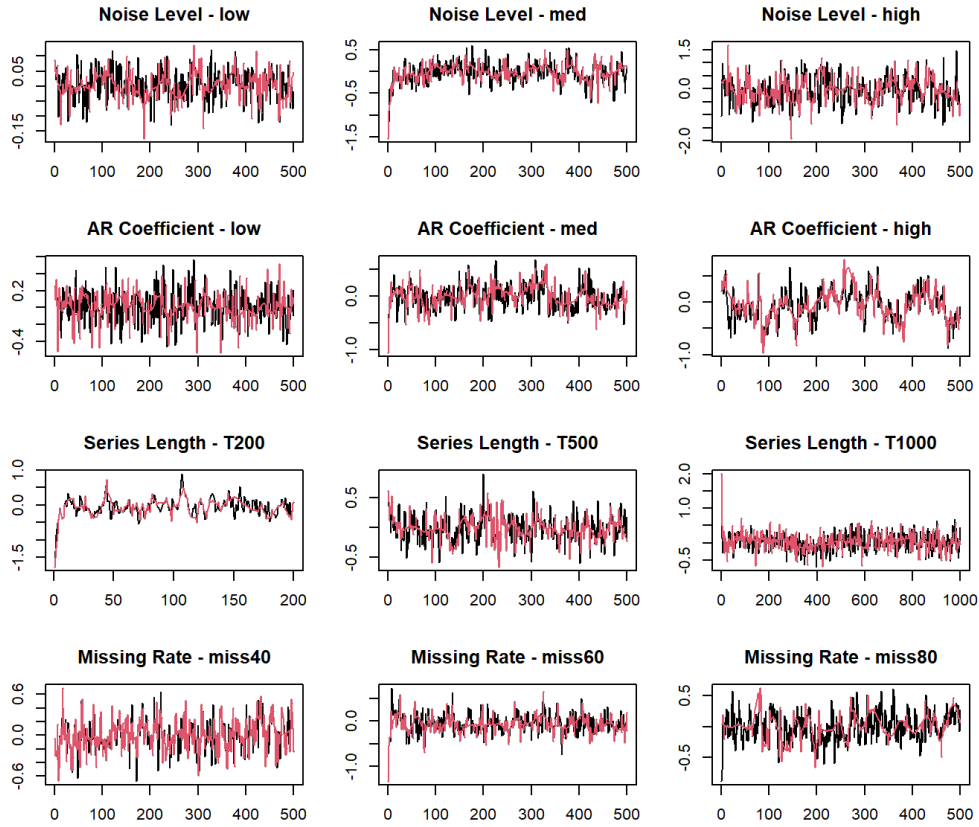


Figure 2: Reconstruction using local polynomial regression under the stationary setting.

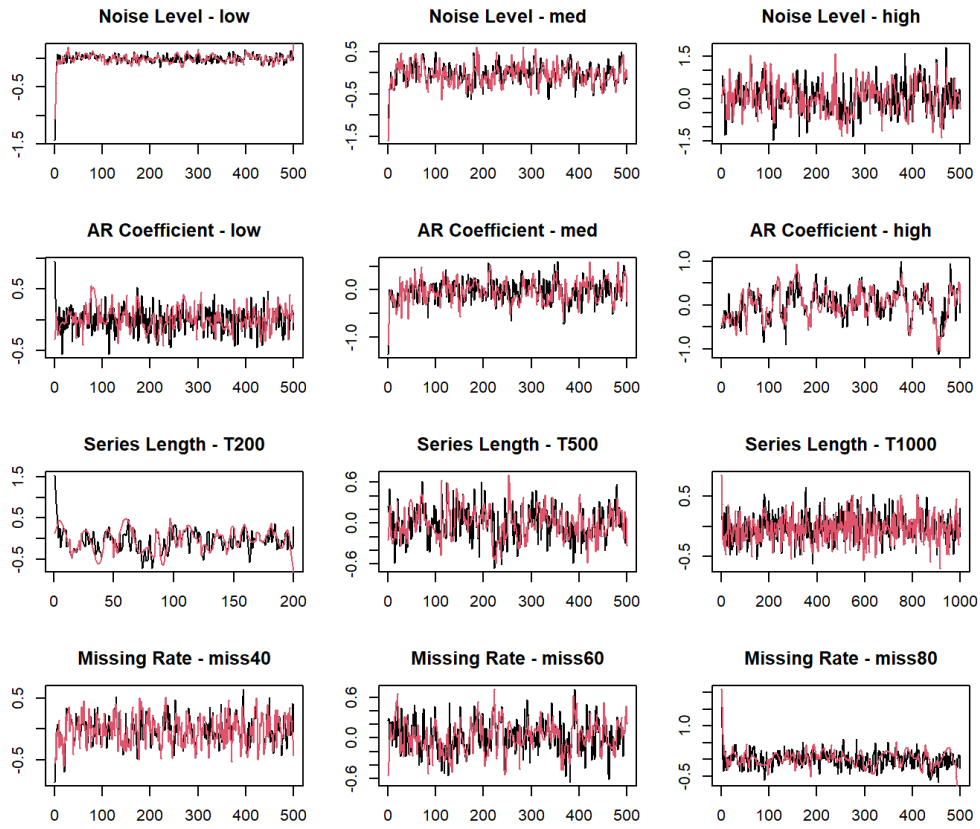


Figure 3: Reconstruction using cubic smoothing splines under the stationary setting.

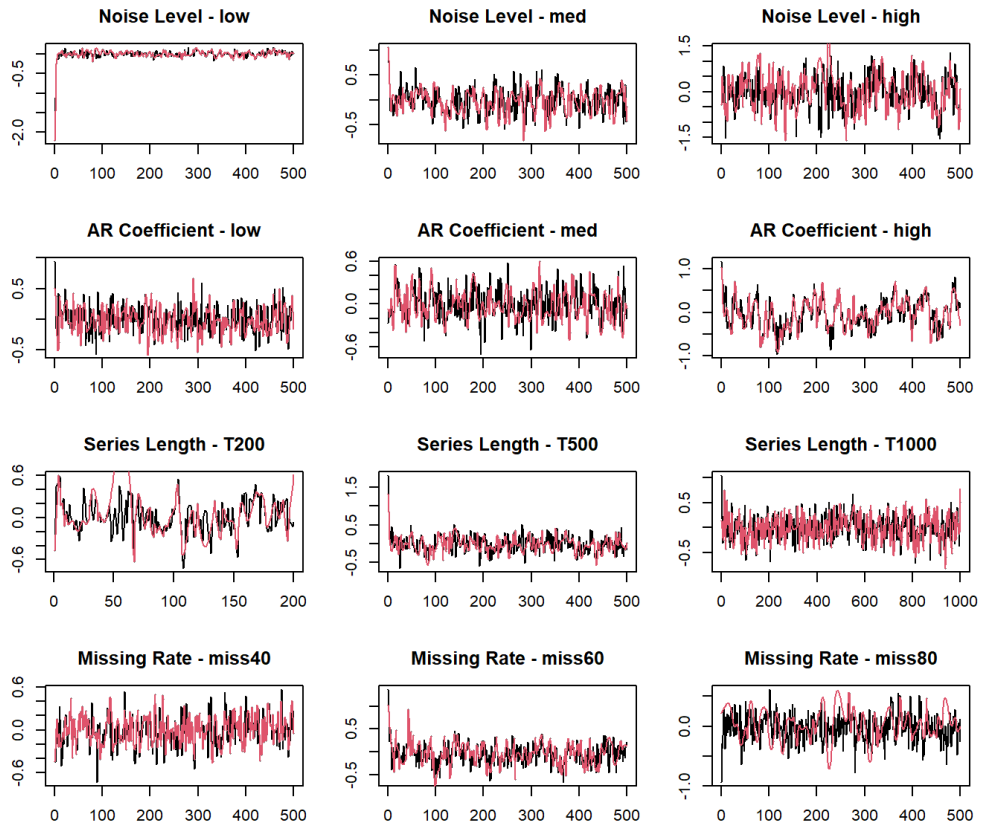


Figure 4: Reconstruction using Kalman filtering under the stationary setting.

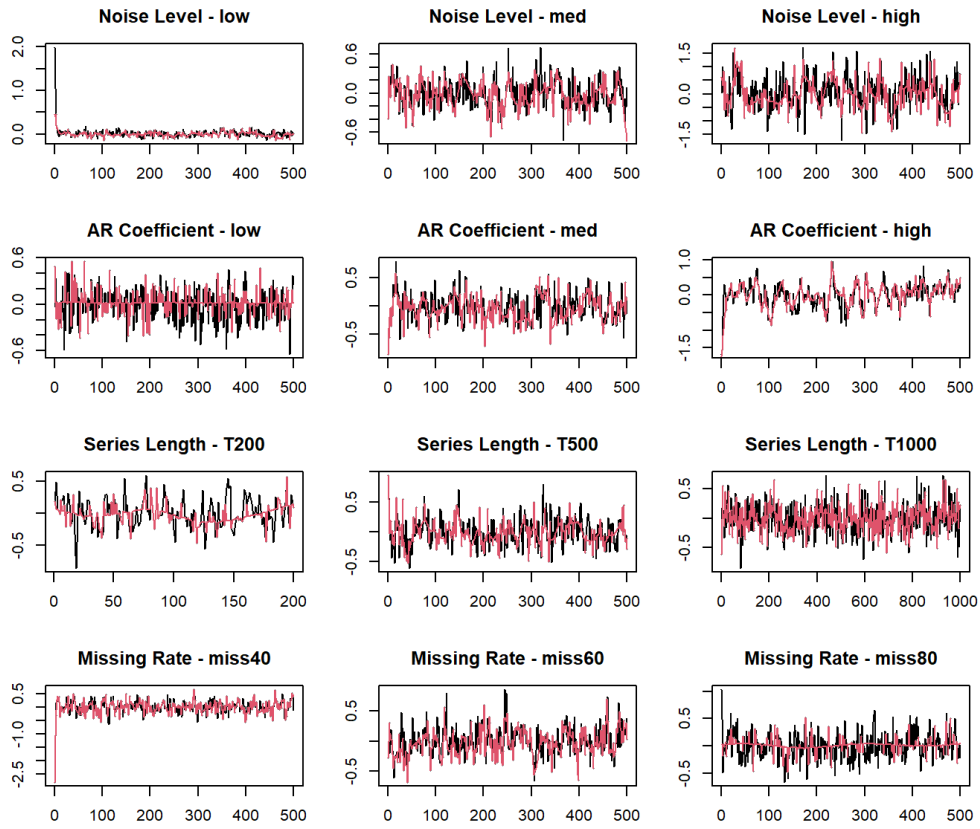


Figure 5: Reconstruction using PACE under the stationary setting.

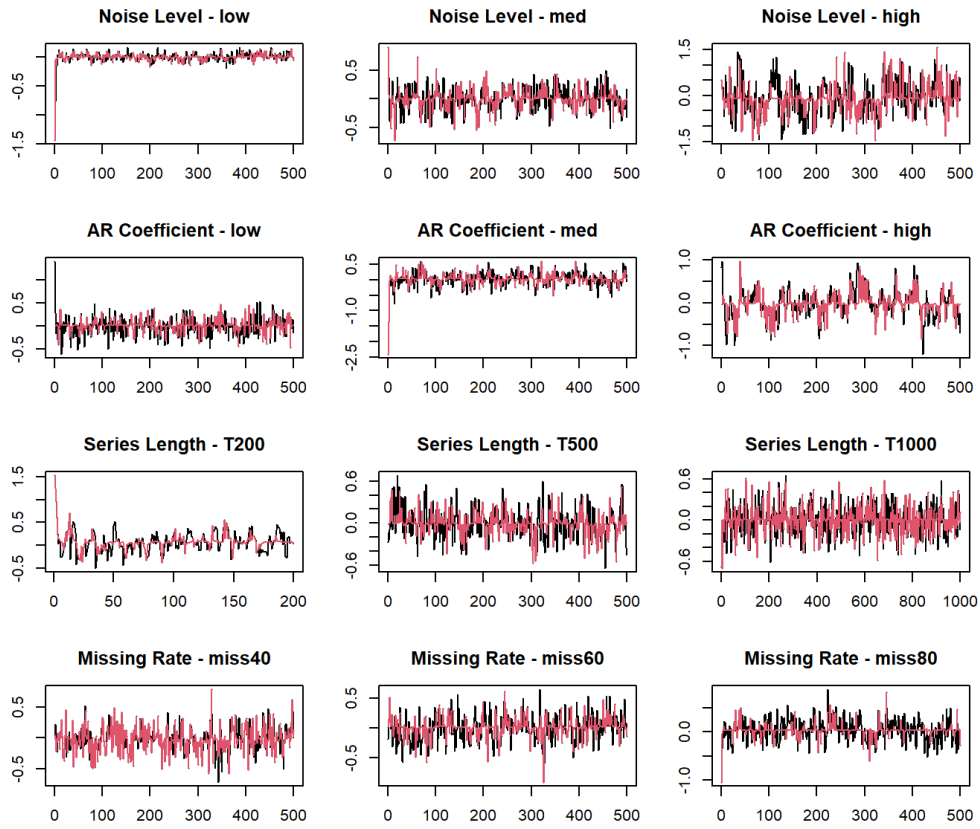


Figure 6: Reconstruction using GRU under the stationary setting.

C.2 Nonstationary Reconstruction Examples

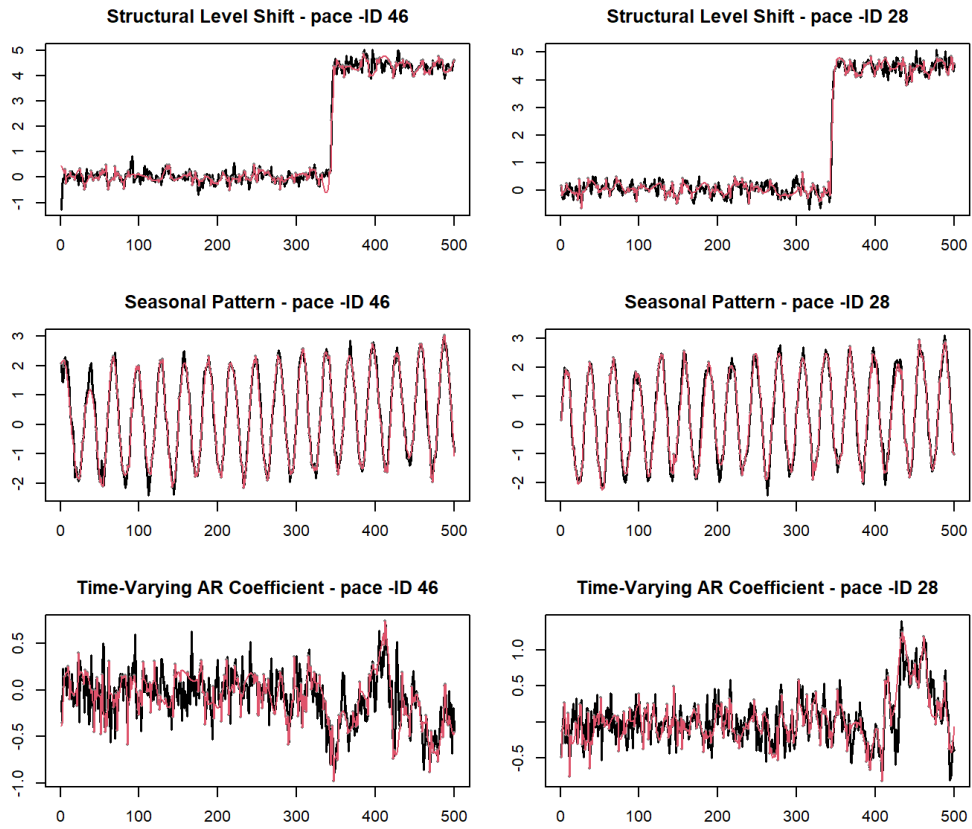


Figure 7: Reconstruction under nonstationary dynamics using PACE.

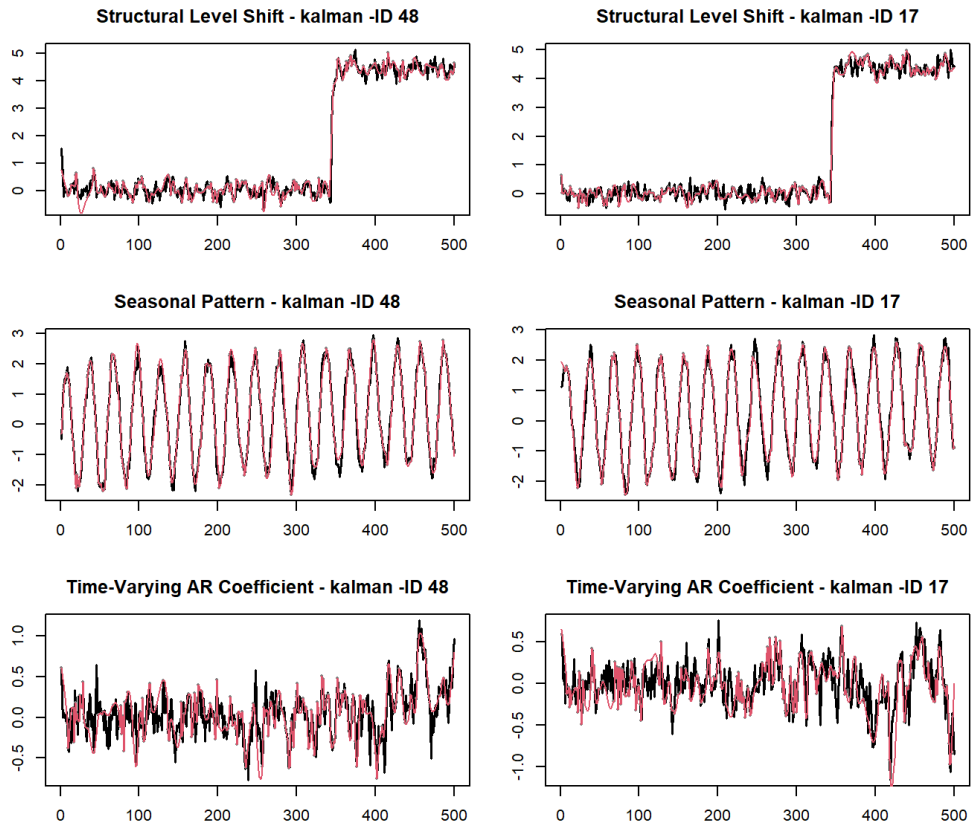


Figure 8: Reconstruction under nonstationary dynamics using Kalman filtering.